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SECURITY INFORMATION AND EVENT MANAGEMENT (SIEM)

NETWORK SECURITY

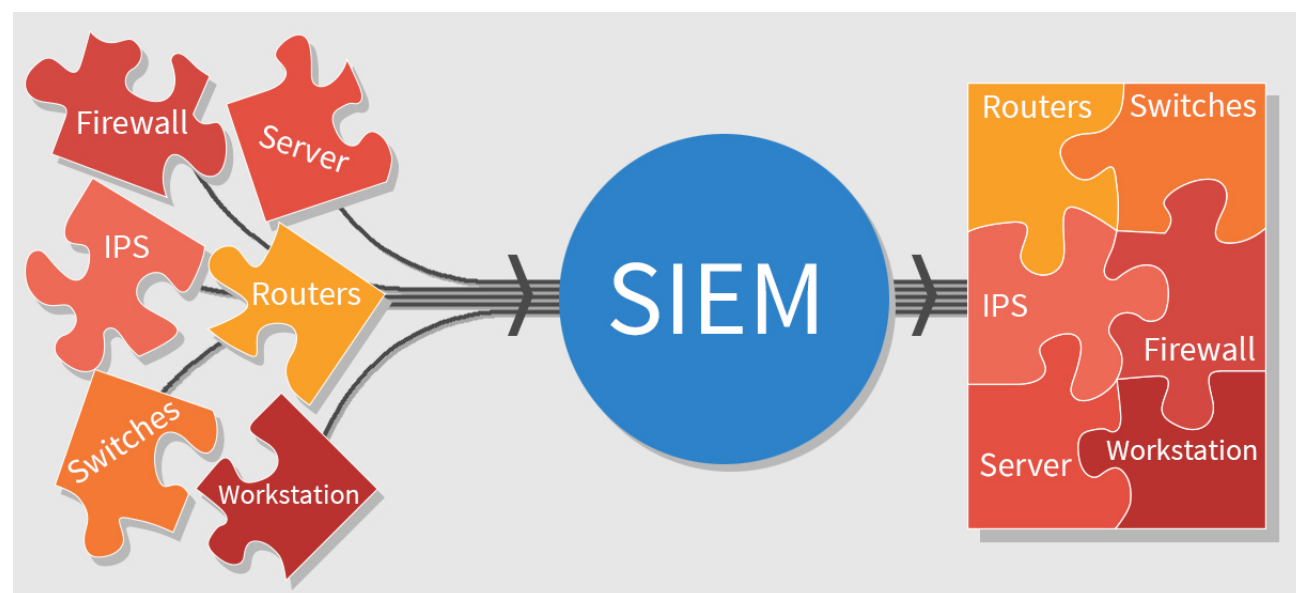
OBJECTIVES

- ▶ SIEM
- ▶ SIEM Main Functions
- ▶ SIEM Key Components
- ▶ SIEM Tools
- ▶ Wazuh



WHAT IS SIEM

- ▶ Security Information and Event Management (SIEM) is a technology that helps organizations monitor and analyze security-related data in real time.
- ▶ Collects and manages logs from various devices, like **servers**, **firewalls**, and **intrusion detection systems**, to provide a centralized view of potential security threats.



SIEM MAIN FUNCTIONS

- ▶ **Log aggregation and management:** Collect logs from multiple sources to get a unified view of security events.
- ▶ **Real-time monitoring and alerting:** Continuously monitors for suspicious activity and trigger alerts.
- ▶ **Event correlation:** Analyze different events to find relationships that might indicate a security incident.
- ▶ **Threat detection:** Identify potential threats from the data collected.
- ▶ **Reporting and compliance:** Generate reports, especially for compliance with regulations like HIPAA or GDPR.

KEY COMPONENTS

- ▶ **Data Collection:** SIEM collects logs and security events from various sources (like firewalls, intrusion detection systems, and servers). This data is the foundation for monitoring and analysis.
- ▶ **Event Correlation:** SIEM looks for patterns across different sources of data.
 - Example, a failed login followed by a file deletion might be correlated to identify a potential security breach.

KEY COMPONENTS CONTINUED

- ▶ **Alerting:** When the SIEM detects something suspicious based on its rules, it generates an alert, which can be reviewed by security analysts.
- ▶ **Dashboards and Reports:** SIEM tools provide visual representations of the data, making it easier for teams to monitor activities and identify security trends.

HOW SIEM WORKS

► Step-by-Step Process:

1. **Log Collection:** SIEM gathers logs and events from various sources (like network devices, servers, and security systems).
2. **Normalization of data:** Data is standardized so that logs from different sources are formatted similarly for easier analysis.
3. **Correlation of events:** Uses rules to detect patterns across different logs (e.g., multiple failed login attempts from the same IP).
4. **Alert generation:** Based on predefined rules, SIEM generates alerts when certain events or patterns are detected.
5. **Incident response and forensic analysis:** SIEM can also be used to analyze past events for deeper investigation.

BENEFITS OF SIEM

- ▶ **Threat Detection:** Detects advanced threats and breaches.
- ▶ **Compliance:** Helps organizations meet regulatory requirements (e.g., GDPR, HIPAA).
- ▶ **Incident Response:** Speeds up detection and mitigation of attacks.
- ▶ **Centralized View:** Provides a holistic view of network and security events.

POPULAR SIEM TOOLS – PROPRIETARY

- ▶ **Splunk Enterprise Security:** A widely used SIEM known for its powerful search capabilities and scalability.
- ▶ **IBM QRadar:** Known for its advanced analytics and real-time monitoring.
- ▶ **ArcSight:** One of the earliest SIEMs, known for large-scale enterprise security monitoring.

POPULAR SIEM TOOLS – OPEN SOURCE

- ▶ **Elastic Stack (ELK):** A popular open-source solution that, when configured, can function as a SIEM. It's known for its flexibility and scalability.
- ▶ **Wazuh:** An open-source SIEM platform that integrates with Elastic Stack for log analysis.
- ▶ **Security Onion:** A security monitoring platform that includes various open-source tools like Suricata (IDS/IPS) and Zeek for network monitoring.
- ▶ **Graylog:** A log management tool that can be extended for security use cases and SIEM capabilities.

REAL-WORLD USE CASES OF SIEM

- ▶ **Detecting Insider Threats:** SIEM can detect unusual patterns, such as an employee accessing sensitive files they usually don't need to, which may indicate a potential insider threat.
- ▶ **Ransomware Detection:** SIEM can detect encryption patterns common in ransomware attacks and trigger alerts before significant damage is done.
- ▶ **DDoS Attack Monitoring:** SIEM helps monitor large-scale traffic patterns that may indicate a Distributed Denial of Service (DDoS) attack, where attackers try to overwhelm a network with traffic.

CHALLENGES WITH SIEM

- ▶ **False Positives:** One of the biggest challenges with SIEM systems is the large number of alerts generated. Many alerts may be false positives, which can overwhelm security teams and reduce effectiveness.
- ▶ **Complexity:** Requires skilled professionals for setup and maintenance.
- ▶ **Cost:** Licensing and data storage costs can be high for large organizations.

FUTURE TRENDS IN SIEM

- ▶ **Machine Learning & AI:** Using AI to improve threat detection and reduce false positives.
- ▶ **Cloud SIEM:** Leveraging cloud infrastructure for scalability.
- ▶ **Integration with SOAR (Security Orchestration, Automation, and Response):** SIEM systems are being integrated with SOAR tools, which automate responses to incidents. This reduces the need for human intervention in routine tasks, freeing up security teams to focus on complex incidents.

WAZUH

- ▶ Wazuh is an open-source SIEM platform that integrates seamlessly with Snort for IDS logs and can also process UFW logs.
- ▶ It has built-in rules for detecting various security threats and offers features like file integrity monitoring, vulnerability detection, and log analysis.
- ▶ Wazuh can collect and analyze logs from **Snort** (both in network IDS and host-based modes) and **UFW**, providing a central dashboard to correlate the alerts and logs.
- ▶ We will implement this next module.

SIEM – ASSIGNMENT 7

- ▶ Reading: <https://www.ibm.com/topics/siem>
- ▶ TryHackMe Lab: <https://tryhackme.com/r/room/introtoiem>
- ▶ Lab on VM: See PDF under Module 7.